Topic 6: Estimation

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Statistics (III) Midterm

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Core Questions

- What makes an estimator a “good” estimator?
- When population parameters are unknown, how do we estimate them from samples?
- Why does the quality of an estimator directly affect SPC?

Why Is Estimation Important in SPC?

In SPC, the in-control state of a process is usually described by baseline parameters such as μ_0, σ_0

In practice, they are usually **unknown** and must be estimated from Phase I data

Key points:

- If $\hat{\mu}_0$ is estimated incorrectly, the center line will shift
- If $\hat{\sigma}_0$ is estimated incorrectly, the control limits will become too wide or too narrow
- Incorrect limits may lead to:
    - Too many false alarms
    - Delayed detection of abnormal changes

Therefore, SPC is not only about drawing charts. The first question is:

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How good is the estimator?

Estimators and Sampling Distributions

Suppose

$$X_1, \ldots, X_n \overset{iid}{\sim} F_\theta$$

where θ is an unknown parameter

An **estimator**is a function of the sample:

$$\hat{\theta} = T(X_1, \ldots, X_n)$$

Since $X_1, \ldots, X_n$ are random, $\hat{\theta}$ is also a random variable
Therefore, we should not only ask "what value do we get?" , but also ask:

- What is $E(\hat{\theta})$?
- What is $\mathrm{Var}(\hat{\theta})$?
- Does $\hat{\theta}$ converge to θ as $n \to \infty$?

Bias: Does the Estimator Deviate from the True Value?

Definition:

$$\mathrm{Bias}(\hat{\theta}, \theta) = E(\hat{\theta}) - \theta$$

If

$$E(\hat{\theta}) = \theta, \quad \forall\theta$$

then $\hat{\theta}$ is called an **unbiased estimator**

Interpretation: Bias measures whether the estimator tends to overestimate or underestimate the true value on average

Example: Suppose $E(X_i) = \mu$ and $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then $E(\bar{X}) = \mu$

Therefore, $\bar{X}$ is an unbiased estimator of μ

MSE: Unbiasedness Is Not Enough

Mean Squared Error is defined as

$$\mathrm{MSE}(\hat{\theta}, \theta) = E\big[(\hat{\theta} - \theta)^2\big]$$

It can be **decomposed** into

$$\boxed{\mathrm{MSE}(\hat{\theta}, \theta) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}^2(\hat{\theta}, \theta)}$$

Key message:

- An estimator can be unbiased but still have very large variance
- A slightly biased estimator may have a much smaller MSE

Example

Goal: Estimate μ, assuming $X_1, \ldots, X_n \sim (\mu, \sigma^2)$

Consider two estimators: $\hat{\mu}_1 = \bar{X}, \quad \hat{\mu}_2 = 0.8\bar{X}$

First estimator: $E(\hat{\mu}_1) = \mu, \quad \mathrm{Var}(\hat{\mu}_1) = \frac{\sigma^2}{n}, \quad$ then $\quad \mathrm{MSE}(\hat{\mu}_1) = \frac{\sigma^2}{n}$

Second estimator:

$$E(\hat{\mu}_2) = 0.8\mu, \quad \mathrm{Var}(\hat{\mu}_2) = 0.8^2\mathrm{Var}(\bar{X}) = 0.64\frac{\sigma^2}{n}$$

$$\mathrm{Bias}(\hat{\mu}_2, \mu) = 0.8\mu - \mu = -0.2\mu,$$

$$\text{Then, } \mathrm{MSE}(\hat{\mu}_2) = 0.64\frac{\sigma^2}{n} + (0.2\mu)^2$$

Conclusion: A biased estimator may still have a smaller MSE if its variance is less

Consistency: Does the Estimator Approach the True Value as the Sample Size Increases?

Definition: If a sequence of estimators $\hat{\theta}_n$ satisfies

$$\hat{\theta}_n \xrightarrow{P} \theta$$

then $\hat{\theta}_n$ is called a **consistent estimator** of θ

That is, $$P(|\hat{\theta}_n - \theta| > \varepsilon) \to 0 \quad (n \to \infty), \quad \forall \varepsilon > 0$$

Interpretation:

- The estimator may have a large error when the sample size is small
- The estimator should become increasingly stable and closer to the true value when the sample size increases

A Common Sufficient Condition: $E(\hat{\theta}_n) \to \theta$ and $\mathrm{Var}(\hat{\theta}_n) \to 0$

Example

Suppose
$$X_1, \ldots, X_n \overset{iid}{\sim} \text{Bernoulli}(p)$$

where $P(X_i = 1) = p, \quad P(X_i = 0) = 1 - p, \quad E(X) = p, \quad Var(X) = p(1-p)$

Since $\hat{p} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \Rightarrow E(\hat{p}) = E(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{1}{n} * n * p = p$

$$\text{Var}(\hat{p}) = \frac{1}{n^2} * n * p(1-p) = \frac{p(1-p)}{n}$$

Therefore
$$E(\hat{p}) \to p \text{ and } \text{Var}(\hat{p}) = \frac{p(1-p)}{n} \to 0 \quad \text{as} \quad n \to \infty$$

So
$$\hat{p} \xrightarrow{P} p \Rightarrow \hat{p} \text{ is consistent}$$

From Estimation to SPC: The Core of Phase I

In Phase I, we first establish an in-control baseline

For an $\bar{X}$-chart, if μ_0 and σ_0 are known, the control limits are

$$UCL = \mu_0 + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \quad LCL = \mu_0 - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}$$

We replace them with estimators $\hat{\mu}_0$ and $\hat{\sigma}_0$

Thus, the actual control limits become

$$\widehat{UCL} = \hat{\mu}_0 + z_{1-\alpha/2}\frac{\hat{\sigma}_0}{\sqrt{n}}, \quad \widehat{LCL} = \hat{\mu}_0 - z_{1-\alpha/2}\frac{\hat{\sigma}_0}{\sqrt{n}}$$

Potential issues:

- If $\hat{\mu}_0$ is biased, the center line shifts.
- If $\hat{\sigma}_0$ has large variance, the control limits become unstable.
- If the estimators are not consistent, we cannot establish a reliable baseline.

How Estimator Affects Monitoring: Example of a p-Chart

Suppose each subgroup has sample size n, and the defect rate is p_0 Then

$$\hat{p} = \frac{X}{n}, \quad X \sim \mathrm{Binomial}(n, p_0)$$

In Phase I, we usually use $\bar{p} = \frac{1}{m}\sum_{i=1}^{m} \hat{p}_i$ to estimate p_0
The control limits of the p-chart are often written as

$$UCL = \bar{p} + z\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}, \quad LCL = \bar{p} - z\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

Conclusion

(1) A good estimator should not be judged only by unbiasedness

- $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$

(2) Consistency is a requirement for long-run reliability

- $\hat{\theta}_n \xrightarrow{P} \theta$

(3) In SPC, estimation is the foundation

- Phase I requires estimation of μ_0 and σ_0 in order to determine control limits
- Control limits determine false alarm rates and detection power

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If the baseline estimation is unreliable,
then the entire SPC monitoring system becomes unreliable.

Thank You